

RACIAL DYNAMICS IN LABOR MARKETS: EXOGENOUS OR ENDOGENOUS?[†]

Black–White Earnings Differentials: Privatization versus Deregulation

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Over 40 years ago Gary Becker (1957) argued that competition helps mitigate the ability of firms to engage in wage and employment discrimination. Deregulated transportation industries, in particular, provide fertile ground to test the Becker hypothesis. The increase in competition from deregulation should make it increasingly costly for employers to exercise discriminatory preferences. Several studies of the deregulated trucking industry provide support for this hypothesis (Nancy L. Rose, 1987; Peoples and Lisa Saunders, 1993; John S. Heywood and Peoples, 1994).

This study also utilizes information on transportation industries to test the Becker hypothesis. It differs from the literature in that one of the transportation industries examined is privatized.¹ At question is whether the Becker hypothesis also holds for a publicly owned transportation industry in a privatized environment. This question is addressed by investigating privatization's effect on black–white earnings differentials of public-transit bus drivers. These results are then compared with deregulation's effect on black–white earnings differentials of private for-hire truck drivers. Such a comparison allows for analyzing differences in differential earnings for comparable occupations in privatization and deregulation regimes.

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¹ The one exception is the study by Peoples and Talley (2000).

I. Transportation Industries

A. Public Transit

An overview of the public-transit industry reveals a business environment that presented nontrivial employment opportunities for blacks before the civil-rights era. Prior to 1960, transit firms were generally privately owned, operating under exclusive franchises subject to government regulation of fares and routes. While such regulation does not necessarily discourage employment discrimination, barriers for blacks as transit bus drivers were relatively low by the 1960's. Successful black employment gains occurred, in part, because the labor-supply schedule for this occupation was (and remains) relatively elastic (Philip W. Jeffress, 1971). The changing demographics of transit riders also contributed to enhancing black transit employment for bus drivers. For instance, by 1960 transit systems serviced a disproportionate share of blacks who resided in densely populated areas. Such a clientele is not likely to object to the employment of minorities. Further, transit unions did not have a policy of restricting black employment. Transit unions were more likely to promote employment opportunities for minorities. The most powerful transit union, the Transport Workers Union of America (TWA), even created its own civil-rights committee to address unfair employment practices.

After 1960 changes in the transit industry's business environment indirectly improved the chances of blacks receiving high wages. The 1960's saw a sharp deterioration in transit ridership, resulting from rising incomes and private automobile use. Declining demand threatened the solvency of many transit firms and precipitated a trend of public takeover and

growing government subsidization. Public-transit firms operating in densely populated areas received a large share of these subsidies. Transit subsidization helped finance higher earnings for workers in this industry, especially blacks who were disproportionately employed as transit workers in densely populated locations.

By 1980, changing attitudes toward the financing of transit services reduced the opportunity for black transit employees to receive high earnings. The Reagan Administration held the view that operating subsidies increased the cost of transit service by financing increases in labor earnings disproportionate to increases in productivity. While the Reagan Administration sought to eliminate federal operating subsidies, resistance from Congress allowed these subsidies to continue, though at significantly lower levels (American Public Transit Association, 1997).

Another effort to lower public-transit costs is depicted by the 1984 transit-privatization initiative of the Urban Mass Transportation Administration (UMTA). This initiative promoted the contracting-out of public-transit service to lower-cost private providers. UMTA's contracting-out privatization policy charged "localities with the responsibility of demonstrating that they were actively encouraging private firms to participate in the provision of new and restructured local services" as a condition for obtaining or retaining matching funds from the federal government (E. D. Sclar et al., 1989 p. 9). Greater emphasis on contracting-out to low-cost private establishments placed downward pressure on the earnings of public-transit workers. Black transit drivers were most vulnerable to this wage pressure, since they were disproportionately employed in public systems that rely on federal subsidies.

B. For-Hire Trucking

Regulation of entry and rates influenced racial employment in the trucking industry, but much differently than the regulation effect in the transit industry. The Motor Carrier Act of 1935 severely restricted entry into the lucrative less-than-truckload (LTL) sector of the for-hire trucking industry. The lack of competition in the highly organized LTL sector depicts a business environment that presents employers and unions with the latitude

to restrict employment of workers from non-preferred groups. Evidence from past research reveals that blacks were one-fourth as likely to be employed in the lucrative segment of the for-hire industry compared to their employment in the rest of the industry (Richard Leone, 1970). Such employment disparities occurred even though truck drivers faced a highly elastic labor-supply curve (Rose, 1987). The lack of black truck drivers employed in the LTL sector was due in part to union seniority rules that relegated transfers to the bottom of the LTL seniority list. Barriers to black LTL employment is particularly significant because rate regulation based on cost mark-up pricing methods allowed payment of high wages to drivers in the LTL sector. The opportunity to negotiate high union wages was further enhanced by the Teamsters' control over the labor supply of LTL truck drivers.

Deregulation initiated by the Interstate Commerce Commission in 1978 and legislated by the 1980 Motor Carrier Act (MCA) profoundly changed the business environment in the for-hire trucking industry. Entry restrictions eased, and federal regulators relied more heavily on the market to set rates. Significant entry of low-cost nonunion carriers occurred following deregulation. This influx of new carriers contributed to the percentage of union drivers in the for-hire truck sector falling from 56.8 percent in 1978 to 24.9 percent in 1995. Trucking deregulation policy also had the unplanned effect of improving employment opportunities for black drivers. The share of for-hire jobs held by black drivers increased from 15.5 percent in 1978 to 24.3 percent in 1995 (Barry T. Hirsch and David A. Macpherson, 1998). Such employment patterns comport well with the view that union and trucking firms are less likely to indulge in employment discrimination when faced with stepped-up competition.

Evidence suggesting that black drivers experienced eroding employment barriers in the still-lucrative LTL sector is not as obvious as for the rest of the for-hire industry. High LTL start-up cost associated with necessary production inputs such as terminals and dock equipment creates a tremendous barrier to greater competition. High LTL barriers to entry contributed to an increasing share of the LTL

freight hauled by the largest carriers in this sector. LTL carriers also remained highly organized. The lack of competition coupled with the union's control over the supply of drivers could still provide employers and unions with the latitude to exercise hiring preferences in the high-wage LTL segment of the for-hire industry. Empirical analysis is then required to examine the extent to which the racial earnings differential declined following deregulation.

II. Data and Earnings Model

Individual information for public-transit bus and private for-hire truck drivers is taken from Current Population Survey (CPS) files of the Bureau of the Census for the years 1973–1996 (excluding 1982) and used to examine black–white earnings differentials in the public-transit and private for-hire trucking industries. Data covering the observation periods from 1973–1981 were taken from May CPS files. Data for the 1983–1996 observation period were taken from 168 monthly CPS outgoing rotation group files for January 1983 through December 1996.² The year 1982 was omitted, because union-status information was not collected in the survey. The selected samples consist of full-time public-transit bus and private for-hire truck drivers, age 16–64, who reported information on their usual weekly earnings, usual hours worked, union status, and a set of demographic characteristics.³

The CPS public-transit bus and private truck driver samples are used to estimate the following earnings equations:

$$\begin{aligned}
 (1a) \quad \ln(\text{Earnings})_{j,t,\text{transit},\text{union}} \\
 = \alpha_1(\text{Black})_{j,t} + \alpha_2(\text{Privatization})_{j,t} \\
 + \alpha_3(\text{Privatization} \times \text{Black})_{j,t} \\
 + \sum \beta_k X_{j,t,k} + \varepsilon_{j,t,\text{transit},\text{union}}
 \end{aligned}$$

² Twelve-month CPS files in which the union question was asked are not available prior to 1983.

³ Truck drivers in the samples are employee drivers. Self-employed truck drivers (i.e., owner-operators) are not considered.

$$\begin{aligned}
 (1b) \quad \ln(\text{Earnings})_{j,t,\text{truck},\text{union}} \\
 = \alpha_1(\text{Black})_{j,t} + \alpha_2(\text{Deregulation})_{j,t} \\
 + \alpha_3(\text{Deregulation} \times \text{Black})_{j,t} \\
 + \sum \beta_k X_{j,t,k} + \varepsilon_{j,t,\text{truck},\text{union}}
 \end{aligned}$$

where j indexes individual drivers, t indexes the observation year, “transit” and “truck” index public-transit bus and private truck sectors, and “union” indexes union status (i.e., union vs. nonunion). The variable $\ln(\text{Earnings})$ is the natural log of usual weekly earnings in 1983–1984 dollars.⁴ The variable “Black” is a dummy equaling 1 if the driver is black. The industrial policy variables “Privatization” and “Deregulation” are dummies equaling 1 for the observation years 1984–1996 and 1979–1996, respectively. The control vector $\mathbf{X}$ includes a constant as well as the standard labor earnings determinants. In addition, the annual national unemployment rate is included as an earnings determinant to control for time-variant distortions such as changes in the business cycle.

The coefficients of particular interest to this study are α_1 and α_3 . The coefficient α_1 measures the black–white log-earnings differential of drivers for the observation period prior to the industrial policy. The coefficient α_3 measures the change in the black–white log-earnings differentials of drivers for the observation period following the introduction of such policy.

III. Estimation Results

Estimates of equation (1a) for union and non-union public transit bus drivers appear in columns (i) and (ii) of Table 1, and estimates of equation (1b) for union and nonunion private for-hire truck drivers appear in columns (iii) and (iv). The estimated coefficient on the “Black” dummy in column (i) indicates an 11.74-percent earnings premium for black union public-transit bus drivers in the pre-privatization period.⁵ This finding is consistent with the view that black

⁴ Usual weekly earnings were deflated using the Consumer Price Index (CPI).

⁵ Earnings differentials are calculated as $(e^\alpha - 1) \times 100$.

TABLE 1—PARTIAL RESULTS ON LOG WEEKLY EARNINGS

Variable	Public-transit bus drivers		Private for-hire truck drivers	
	Union (i)	Nonunion (ii)	Union (iii)	Nonunion (iv)
Black	0.111 (2.552)	-0.0589 (-0.661)	-0.155 (-3.418)	-0.240 (-3.717)
Privatization	0.045 (1.293)	0.082 (1.296)	—	—
Privatization × Black	-0.1407 (-2.29)	0.1138 (1.010)	—	—
Deregulation	—	—	-0.155 (-10.769)	-0.129 (-6.444)
Deregulation × Black	—	—	0.062 (1.291)	0.225 (3.403)
Sample size:	737	327	5,326	10,465

Notes: The control variables include age, age-squared, log of weekly hours worked, annual U.S. unemployment rate, and dummies for marital status, gender, five levels of educational attainment, and residence in three of the four U.S. geographic quadrants. Numbers in parentheses are *t* statistics. Complete results are available from the authors upon request.

union transit drivers benefited from the pre-privatization transit labor environment. This finding is also unique to union transit drivers in that the estimated coefficient on the “Black” dummy in column (ii) indicates that black non-union transit drivers received earnings 5.72-percent below those of their white counterparts in the pre-privatization period.

The results in column (i) support the Becker hypothesis that stepped-up competition is associated with smaller racial earnings differentials. For example, the estimated coefficient on the Privatization × Black interaction term indicates a 13.12-percent erosion of the black union earnings premium for the period. This erosion left black union transit drivers with earnings closely resembling those of their white counterparts. Further, the earnings disadvantage of black nonunion transit drivers eroded in the post-privatization period, with their earnings slightly exceeding those of their white counterparts.

Findings in column (iii) indicate that black union truck drivers received an earnings discount below that of their white counterparts in the pre-deregulation period. The estimated coefficient on the “Black” variable indicates that black union truck drivers received a 14.35-

percent discount in earnings. Findings in column (iv) indicate an even larger black discount for nonunion drivers. The estimated coefficient on the “Black” variable indicates that black nonunion truck drivers received earnings that were 21.33 percent below those of their white counterparts.

The estimated coefficient on the Deregulation × Black interaction term in column (iii) indicates a 6.4-percent-point erosion of the earnings discount for black union truck driver earnings in the post-deregulation period. While this result supports the Becker hypothesis, the remaining 7.95-percent earnings advantage for white truck drivers is also consistent with the notion that a racial earnings differential can still persist without stepped-up competition in all segments of the for-hire trucking industry. The results in column (iv) indicate a complete erosion of the earnings disadvantage for black non-union truck drivers in the post-deregulation period.

IV. Conclusion

Industrial policy over the last 25 years has emphasized greater reliance on competitive markets to enhance consumer welfare. Such policy may also have the unintended effect of reducing racial wage disparities. Support for this view has been reported for the deregulated trucking industry. This study finds that privatization in the public transit sector is associated with declines in the racial earnings differential that resemble those found in trucking. The privatization results are unique in that they reveal that it is the earnings advantage of blacks that erodes following the introduction of this policy. In sum, the results from this study suggest that Becker’s hypothesis on competition and discrimination holds for publicly owned establishments in a privatized environment, at least when the occupation used for analysis requires modest training.

REFERENCES

- American Public Transit Association.** *Transit fact book*. Washington, DC: American Public Transit Association, 1997.
- Becker, Gary S.** *The economics of discrimination*. Chicago: University of Chicago Press, 1957.

- Heywood, John S. and Peoples, James, Jr.** "Deregulation and the Prevalence of Black Truck Drivers." *Journal of Law and Economics*, April 1994, 37(1), pp. 133–55.
- Hirsch, Barry T. and Macpherson, David A.** "Earnings and Employment in Trucking: Deregulating a Naturally Competitive Industry," in J. Peoples, ed., *Regulatory reform and labor markets*. Boston, MA: Kluwer, 1998, pp. 61–112.
- Jeffress, Philip W.** "The Negro in the Urban Transit Industry," in H. R. Northrup, H. W. Risher, Jr., R. D. Leone, and P. W. Jeffress, eds., *Negro employment in land and air transport*. Philadelphia: University of Pennsylvania Press, 1971, pp. 1–101.
- Leone, Richard.** *The Negro in the trucking industry*. Philadelphia: University of Pennsylvania Press, 1970.
- Peoples, James, Jr. and Saunders, Lisa.** "Trucking Deregulation and the Black/White Wage Gap." *Industrial and Labor Relations Review*, October 1993, 47(1), pp. 23–35.
- Peoples, James, Jr. and Talley, Wayne K.** "Privatization, City Residency, and Black–White Earnings Differentials: Evidence from the Public Transit Sector." Unpublished manuscript, University of Wisconsin–Milwaukee, 2000.
- Rose, Nancy L.** "Labor Rent Sharing and Regulation: Evidence from the Trucking Industry." *Journal of Political Economy*, December 1987, 95(6), pp. 1146–78.
- Sclar, E. D.; Schaeffer, K. H. and Brandwein, R.** *The emperor's new clothes: Transit privatization and public policy*. Washington, DC: Economic Policy Institute, 1989.

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3. Sarah Brown, Daniel Gray, Alberto Montagnoli. 2019. Credit supply shocks and household leverage: Evidence from the US banking deregulation. *Journal of Financial Stability* **43**, 97-115. [[Crossref](#)]
4. Benjamin Bental, Vered Kraus, Yuval Yonay. 2017. Ethnic and gender earning gaps in a liberalized economy: The case of Israel. *Social Science Research* **63**, 209-226. [[Crossref](#)]
5. Clémence Berson. 2016. Local labor markets and taste-based discrimination. *IZA Journal of Labor Economics* **5**:1. . [[Crossref](#)]
6. Blaise Melly, Patrick A. Puhani. 2013. DO PUBLIC OWNERSHIP AND LACK OF COMPETITION MATTER FOR WAGES AND EMPLOYMENT? EVIDENCE FROM PERSONNEL RECORDS OF A PRIVATIZED FIRM. *Journal of the European Economic Association* **11**:4, 918-944. [[Crossref](#)]
7. Adelaide Figueiredo, Fernanda Figueiredo, Natália P. Monteiro, Odd Rune Straume. 2012. Restructuring in privatised firms: A Statis approach. *Structural Change and Economic Dynamics* **23**:1, 108-116. [[Crossref](#)]
8. Azim Essaji, Gregory Sweeney, Alexandros Kotsopoulos. 2010. Equality through exposure to imports? International trade and the racial wage gap. *Journal of Economic Policy Reform* **13**:4, 313-323. [[Crossref](#)]
9. Reginald A. Byron. 2010. Discrimination, Complexity, and the Public/Private Sector Question. *Work and Occupations* **37**:4, 435-475. [[Crossref](#)]
10. Natália Pimenta Monteiro. 2010. Using propensity matching estimators to evaluate the impact of privatization on wages. *Applied Economics* **42**:10, 1293-1313. [[Crossref](#)]
11. Ross Levine, Alexey Levkov, Yona Rubinstein. 2009. Racial Discrimination and Competition. *SSRN Electronic Journal* . [[Crossref](#)]
12. Jacqueline Agesa, Kristen Monaco. 2004. Industry racial employment by skill level: The effects of market structure and racial wage gaps. *Journal of Labor Research* **25**:2, 315-328. [[Crossref](#)]
13. Gunseli Berik, Yana van der Meulen Rodgers, Joseph E. Zveglic. 2004. International Trade and Gender Wage Discrimination: Evidence from East Asia. *Review of Development Economics* **8**:2, 237-254. [[Crossref](#)]